



SYDNEY BOYS HIGH  
MOORE PARK, SURRY HILLS

**JUNE 2007**  
**TASK #2**  
**YEAR 12**

## Mathematics Extension 2

### General Instructions:

- Reading time—5 minutes.
- Working time—2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- Start each NEW section in a separate answer booklet.
- Hand in your answer booklets in 3 sections:  
Section A(Questions 1 and 2),  
Section B(Questions 3 and 4),  
Section C(Questions 5 and 6),

### Total marks—90 Marks

- Attempt questions 1–6.
- All questions are of equal value.

**Examiner:** Mr D. Hespe

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

## Section A

Marks

### Question 1 (15 marks)

(a) Find  $\int x \sec^2(1 - x^2) dx$ . 2

(b) Find  $\int \frac{x^2 - 4x + 5}{x - 2} dx$ . 2

(c) (i) Find the values of constants  $A$ ,  $B$ , and  $C$  if 2

$$\frac{A}{x+1} + \frac{Bx+C}{x^2+1} = \frac{4x^2-2x}{(x+1)(x^2+1)}.$$

(ii) Hence or otherwise find  $\int \frac{4x^2-2x}{(x+1)(x^2+1)} dx$ . 3

(d) Given that  $\int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin x + \cos x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos^3 x}{\sin x + \cos x} dx$ , 3

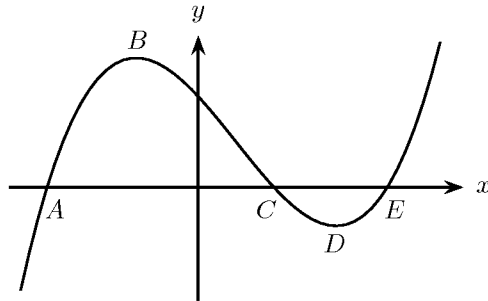
evaluate  $\int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin x + \cos x} dx$ .

(e) Use the method of integration by parts to evaluate  $\int_0^{\ln 2} xe^{-2x} dx$ . 3

**Question 2 (15 marks)**

- (a) A sketch of
- $y = f'(x)$
- is shown below:

3



Draw a neat sketch of a *possible* representation of  $f(x)$ . Mark on your diagram the points  $A'$ ,  $B'$ ,  $C'$ ,  $D'$ , and  $E'$  which are related to the points  $A$ ,  $B$ ,  $C$ ,  $D$ , and  $E$ .

- (b) (i) Draw a neat sketch of  $f(x) = (x + 1)(2 - x)(x - 4)$  showing the intercepts on the axes.

2

- (ii) Without using calculus, use your graph of  $f(x) = (x + 1)(2 - x)(x - 4)$  to sketch, on separate axes, the graphs of:

( $\alpha$ )  $f(x) = |(x + 1)(2 - x)(x - 4)|$

1

( $\beta$ )  $f(x) = \frac{1}{(x + 1)(2 - x)(x - 4)}$

2

( $\gamma$ )  $f(x) = \sqrt{(x + 1)(2 - x)(x - 4)}$

2

- (c) Draw a neat sketch of  $y = \sin x \sin 2x$  for  $-2\pi \leq x \leq 2\pi$  by considering

5

- (i) the zeroes of the function,
- (ii) the sign of the function in different sections of its domain,
- (iii) whether the function is odd or even.

## Section B

(Use a separate writing booklet.)

Marks

### Question 3 (15 marks)

(a) Given that  $z = -2 + 2i$  and  $w = -1 - \sqrt{3}i$ ,

(i) find

( $\alpha$ )  $z - w$ ,

1

( $\beta$ )  $\Re(z^2)$ ,

1

( $\gamma$ )  $|w|$ ,

1

( $\delta$ )  $z\bar{w}$ .

1

(ii) Write  $z$ ,  $\bar{w}$ , and  $\frac{z}{\bar{w}}$  in modulus and argument form.

3

(iii) Hence determine the number  $\lambda$  which satisfies

2

$$\lambda \left( \frac{z}{\bar{w}} \right) = \left( \frac{z}{\bar{w}} \right)^{25}$$

.

(b) If the argument of the complex number  $(z - 1)/(z + 1)$  is  $\frac{1}{4}\pi$  show that  $z$  lies upon a fixed circle whose centre is at the point which represents  $i$ .

3

(c) Find a complex number  $z$  whose argument is  $\frac{1}{6}\pi$  such that

3

$$|z - \sqrt{3} - i| = |z - 2\sqrt{3} - 2i|$$

**Question 4 (15 marks)**

(a) Factorise  $z^4 + z^2 - 6$  over

(i)  $\mathbb{R}$

1

(ii)  $\mathbb{C}$

1

(b) If  $\alpha, \beta, \gamma$  are roots of the equation  $2x^3 - 4x^2 - 6x + 5 = 0$ , find the cubic equation with roots  $\alpha - 1, \beta - 1, \gamma - 1$ .

2

(c) It is suspected that  $12x^3 - 4x^2 - 5x + 2 = 0$  has a repeated root. Use this to find all the solutions.

4

(d) (i) The roots of the quadratic polynomial  $x^2 + px + q = 0$  (where  $q \neq 0$ ) are  $\alpha$  and  $\beta$ , and one root of the quadratic equation  $x^2 + p'x + q = 0$  is  $k\alpha$ . Show that the other root of this equation is  $\frac{\beta}{k}$ .

2

(ii) Assuming that  $k^2 \neq 1$ , write down for each equation an expression for the sum of its roots and hence find  $\alpha$  and  $\beta$  in terms of  $p, p'$ , and  $k$ .

2

(iii) Deduce that  $k(kp - p')(kp' - p) = (k^2 - 1)^2 q$ .

1

(iv) For any given values of  $p, p'$ , and  $q$ , show that the sum of the four possible values of  $k$  is  $\frac{pp'}{q}$ .

2

## Section C

(Use a separate writing booklet.)

Marks

### Question 5 (15 marks)

- (a) If no two boys are to sit together, in how many ways can six girls and three boys be arranged
- (i) in a row, 2
- (ii) around a circular table? 1
- (b) A pair of integers is selected at random from the set of positive integers  $1, 2, 3, \dots, n$ .
- (i) In how many ways may the selection be made? 1
- (ii) If  $n$  is an odd integer, show that the probability that both integers in the pair selected will be odd is  $\frac{n+1}{4n}$ . 2
- (iii) If  $n$  is even, find the probability that both integers in the pair are odd. 2
- (c) Establish a reduction formula for  $\mathbf{I}_n = \int \sec^n x \, dx$ . 2
- (d) (i) Any integer  $n$  can be expressed in one of the forms  $4p, 4p+1, 4p+2, 4p+3$ , where  $p$  is an integer. Find, for each of these forms, the value of the expression  $(1+i^n)(1+i^{2n})$  where  $i^2 = -1$ . 4
- (ii) Hence write down a function of  $n$  which has the value 366 when  $n$  is a multiple of 4 and 365 for all other integral values of  $n$  1

**Question 6 (15 marks)**

(a)  $\int_0^2 \sqrt{4+x^2} dx$  2

[NOTE: you may find the result from Q5(c) useful.]

(b) (i) Determine how the number of stationary points on the curve  $y = e^{-x^2}(x^2+c)$  depends on the value of the real constant  $c$ . 2

(ii) Find the coördinates of the points of inflexion on the curve for which  $c = 2$ . 3

(iii) Sketch the curve for which  $c = 2$ . 2

(c) A rock of mass 5 kg is projected vertically upwards into the air from the ground with initial speed of  $12 \text{ ms}^{-1}$ . The rock is subject to air resistance of  $\frac{v^2}{2}$  Newtons in the opposite direction to its velocity,  $v \text{ ms}^{-1}$ . The rock is also subject to a downward gravitational force of 50 Newtons.

(i) Find the greatest height reached by the rock. 2

(ii) Find the time taken to reach the highest point. 2

(iii) Find how fast the rock is travelling when it hits the ground. 2

**End of Paper**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x$ ,  $x > 0$

# QUESTION 1

$$(a) \int x \sec^2(1-x^2) dx \quad 2$$

$$= -\frac{1}{2} \tan(1-x^2) + C$$

$$(b) \int \frac{x^2 - 4x + 5}{x-2} dx \quad 2$$

$$= \int \left( x - 2 + \frac{1}{x-2} \right) dx \quad ①$$

$$= x^2 - 2x + \frac{1}{2} \ln|x-2| + C$$

(c)  
(i)  $4x^2 - 2x = A(x^2+1) + (Bx+C)(x+1)$   
 $A=3 \quad B=1 \quad C=-3 \quad 2$

$$\therefore I = \int \left( \frac{3}{x+1} + \frac{x-3}{x^2+1} \right) dx \quad 3$$

$$= 3 \ln|x+1| + \frac{1}{2} \ln(x^2+1) - 3 \tan^{-1} x + C$$

$$(d) I = \frac{1}{2} \int \frac{\sin^3 x + \cos^3 x}{\sin x + \cos x} dx \quad 3$$

$$= \frac{1}{2} \int \frac{\sin^2 x + \cos^2 x - \sin x \cos x}{\sin x + \cos x} dx$$

$$= \frac{1}{2} \int \left( 1 - \frac{1}{2} \sin 2x \right) dx \quad ②$$

$$= \frac{1}{2} \left( x - \frac{1}{2} \cdot -\cos 2x \cdot \frac{1}{2} \right)$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{s^3}{s^2+c} dx = \left[ \frac{x}{2} + \frac{1}{8} \cos 2x \right]_0^{\frac{\pi}{2}} = \frac{\pi-1}{4}$$

$$(e) \int_0^{\ln 2} x e^{-2x} dx$$

$$= \int_0^{\ln 2} x \cdot \frac{d}{dx} \left( \frac{1}{2} e^{-2x} \right) dx \quad 3$$

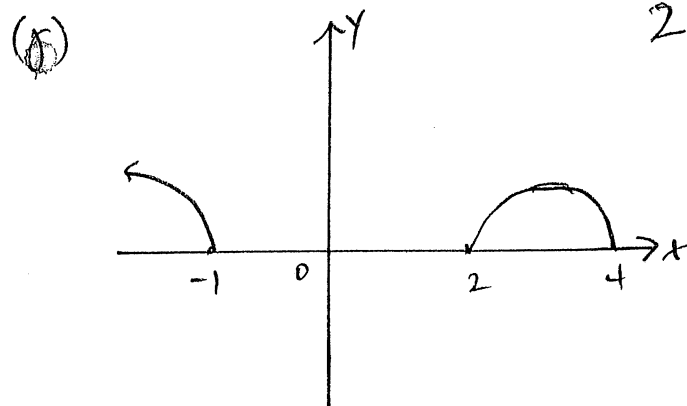
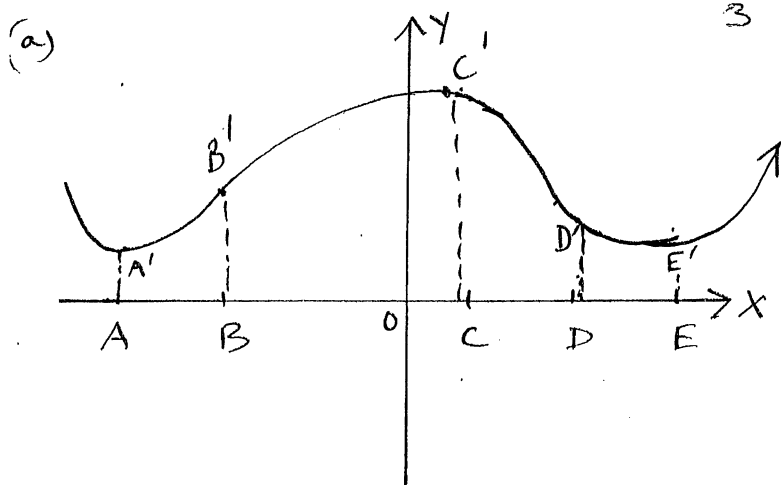
$$= \left[ -\frac{1}{2} x \cdot e^{-2x} \right]_0^{\ln 2} - \int_0^{\ln 2} -\frac{1}{2} \cdot e^{-2x} \cdot 1 dx \quad ④$$

$$= \left[ -\frac{\ln 2}{2} e^{-2 \ln 2} \right] + \left[ -\frac{1}{4} \right] \left[ e^{-2 \ln 2} - e^0 \right]$$

$$= -\frac{\ln 2}{2} \cdot 2^{-2} + \left( -\frac{1}{4} \right) (2^{-2} - 1)$$

$$= -\frac{1}{8} \ln 2 + \frac{3}{16}$$

## QUESTION 2



(c)  $\sin x \sin 2x = 0 \Rightarrow 5$

$\sin x = 0$

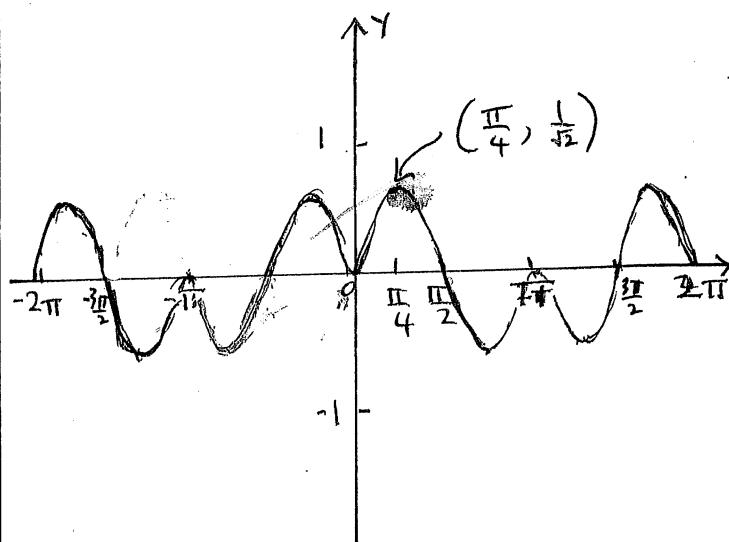
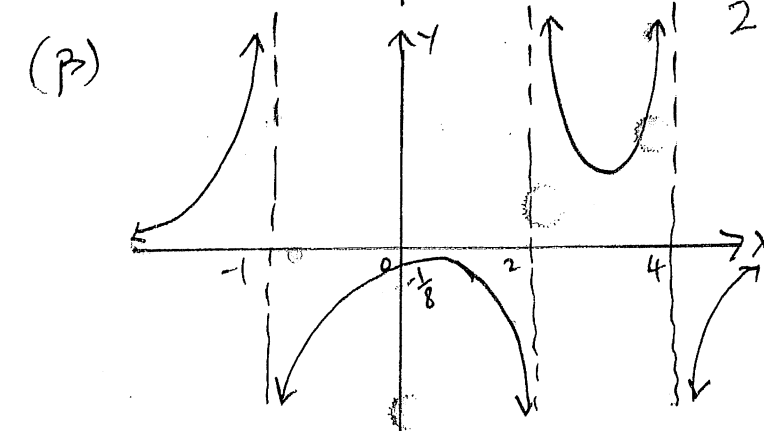
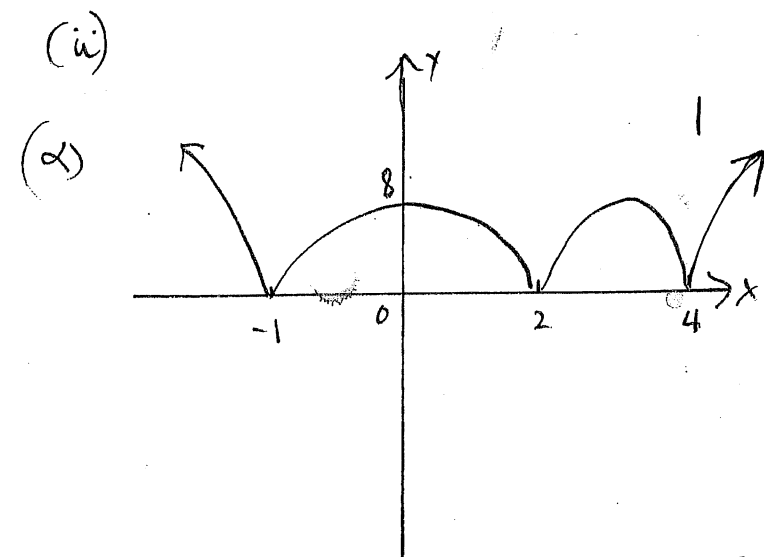
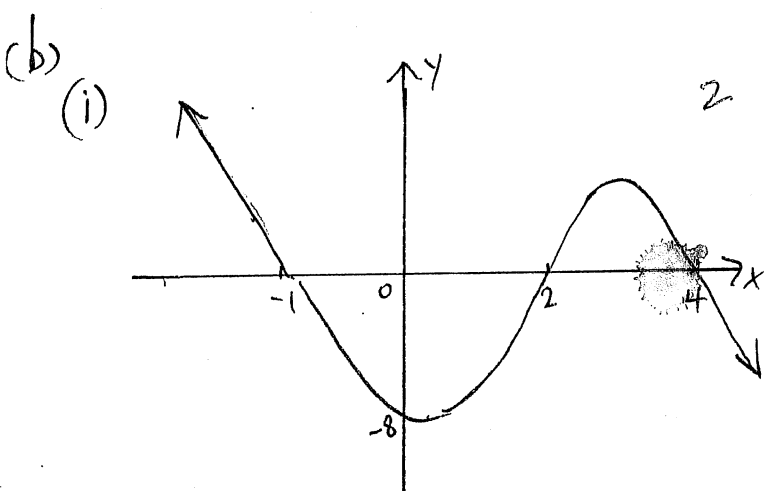
$x = -2\pi, -\pi, 0, \pi, 2\pi$

OR

$\sin 2x = 0$

i.e.  $2x = -4\pi, -3\pi, -2\pi, -\pi, 0, \pi, 2\pi, 3\pi, 4\pi$

$\therefore x = -2\pi, -\frac{3\pi}{2}, -\pi, -\frac{\pi}{2}, 0, \pi, \frac{3\pi}{2}, 2\pi$



$f(x) = \sin x \sin 2x$

$f(-x) = \sin(-x) \cdot \sin(-2x)$

$= -\sin x \cdot -\sin 2x$

$= \sin x \cdot \sin 2x$

$= f(x) \therefore f(x) \text{ even}$

## Section B Question 3

(a)  $z = -2 + 2i$   $w = -1 - \sqrt{3}i$

(i) (a)  $z - w = (-2 + 2i) - (-1 - \sqrt{3}i)$   
 $= -1 + (2 + \sqrt{3})i$  [1]

(b)  $\operatorname{Re}(z^2) = \operatorname{Re}(-2 + 2i)^2$   
 $= \operatorname{Re}(-8i)$   
 $= 0$  [1]

(c)  $|w| = \sqrt{1^2 + \sqrt{3}^2}$   
 $= 2$  [1]

(d)  $z\bar{w} = (-2 + 2i)(-1 + \sqrt{3}i)$   
 $= 2 - 2\sqrt{3}i - 2i - 2\sqrt{3}$   
 $= (2 - 2\sqrt{3}) - (2 + 2\sqrt{3})i$  [1]

(ii)  $z = -2 + 2i$   $\arg z = \frac{3\pi}{4}$   
 $|z| = 2\sqrt{2}$

$\therefore z = 2\sqrt{2} \operatorname{cis} \frac{3\pi}{4}$  [1]

$\bar{w} = -1 + \sqrt{3}i$   $\arg \bar{w} = \frac{2\pi}{3}$   
 $|\bar{w}| = 2$

$\therefore \bar{w} = 2 \operatorname{cis} \frac{2\pi}{3}$  [1]

$\frac{z}{\bar{w}} = \frac{2\sqrt{2} \operatorname{cis} \frac{3\pi}{4}}{2 \operatorname{cis} (\frac{2\pi}{3})}$   
 $= \sqrt{2} \operatorname{cis} (\frac{3\pi}{4} - \frac{2\pi}{3})$   
 $= \sqrt{2} \operatorname{cis} \frac{\pi}{12}$  [1]

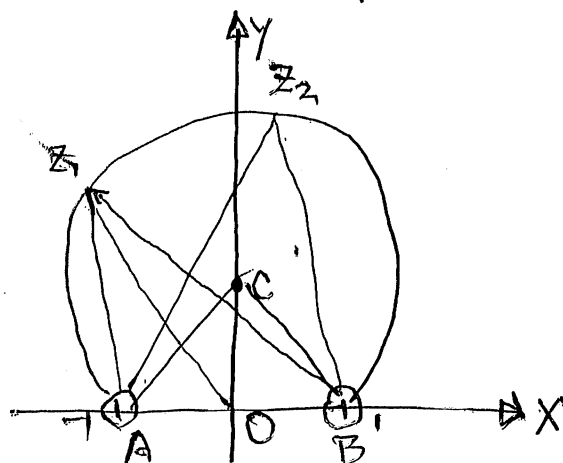
(iii)  $\lambda \left( \frac{z}{\bar{w}} \right) = \left( \frac{z}{\bar{w}} \right)^{25}$   
 $\therefore \lambda = \left( \frac{z}{\bar{w}} \right)^{24}$

$= (\sqrt{2})^{24} \operatorname{cis} \left( \frac{\pi}{12} \times 24 \right)$

$= 4096 \operatorname{cis} 2\pi$

$= 4096$  [2]

(b)  $\arg \left( \frac{z-1}{z+1} \right) = \frac{\pi}{4}$



Let  $Z_1, Z_2$  be points representing two values of  $z$ . Now  $\angle AZ_1B = \angle AZ_2B = \frac{\pi}{4}$

$\therefore$  By the theorem, all  $z$  lie on a circle.

Let  $C$  be the centre of the circle.

$\therefore \angle ACB = \frac{\pi}{2}$  (angle at the centre twice angle at circumference).

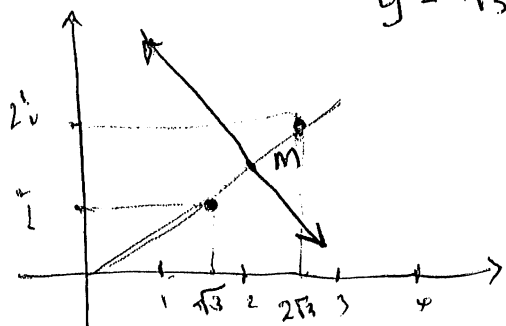
$\therefore$  By symmetry and similarity,  $OC = 1$

$\therefore$  Centre is  $i$ . [3]

$$(c) \arg z = \frac{\pi}{6}$$

$$|z - (\sqrt{3} + i)| = |z - 2(\sqrt{3} + i)|$$

This is the perp. bisector of the interval joining  $\sqrt{3} + i$  and  $2\sqrt{3} + 2i$   
 $y = -\sqrt{3}x + 3$



$$\text{Midpoint } M \text{ is } \frac{3\sqrt{3}}{2} + \frac{3i}{2}$$

We seek  $z$  lying on this line with  $\arg z = \frac{\pi}{6}$ .

$$\text{But } \arg(M) = \frac{\pi}{6}$$

$$\therefore z = M$$

$$= \frac{3}{2}(\sqrt{3} + i)$$

$$= 3 \cos \frac{\pi}{6}$$

Question 4

$$(a) z^4 + z^2 - 6$$

$$= (z^2 + 3)(z^2 - 2)$$

$$(i) = (z^2 + 3)(z + \sqrt{3}i)(z - \sqrt{3}i) \quad \boxed{1}$$

$$(ii) = (z + \sqrt{3}i)(z - \sqrt{3}i)(z + \sqrt{2})(z - \sqrt{2}) \quad \boxed{1}$$

$$(b) 2x^3 - 4x^2 - 6x + 5 = 0 \text{ has roots } \alpha, \beta, \gamma.$$

$$\text{Let } y = x - 1$$

$$\therefore x = y + 1$$

$$\text{Thus } 2(y+1)^3 - 4(y+1)^2 - 6(y+1) + 5 = 0$$

$$\text{has roots } \alpha-1, \beta-1, \gamma-1$$

Simplifying:

$$2(y^3 + 3y^2 + 3y + 1) - 4(y^2 + 2y + 1)$$

$$- 6y - 6 + 5 = 0$$

$$2y^3 + 6y^2 + 6y + 2 - 4y^2 - 8y - 4$$

$$- 6y - 1 = 0$$

$$2y^3 + 2y^2 - 8y - 3 = 0$$

Replace  $y$  with  $x$ :

$$2x^3 + 2x^2 - 8x - 3 = 0 \quad \boxed{2}$$

has real roots.

$$(c) 12x^3 - 4x^2 - 5x + 2 = 0 \quad \text{--- ①}$$

Derived eq'n:

$$36x^2 - 8x - 5 = 0 \quad \text{--- ②}$$

$$x = \frac{8 \pm \sqrt{64 + 4 \times 180}}{72}$$

$$= \frac{1}{2}, -\frac{5}{18}$$

Test  $x = \frac{1}{2}$  in Eq'n ①

$$12\left(\frac{1}{2}\right)^3 - 4\left(\frac{1}{2}\right)^2 - 5\left(\frac{1}{2}\right) + 2 = 0$$

$\therefore x = \frac{1}{2}$  is a repeated root.  $\boxed{1}$

$\therefore (x - \frac{1}{2})^2$  is a factor of ①.

$$x^2 - 1 + \frac{1}{4} \overline{12x^3 - 4x^2 - 5x + 2}$$

$$12x^3 - 12x^2 + 3x$$

$$8x^2 - 8x + 2$$

$$8x^2 - 8x + 2$$

0

$\therefore$  Roots are  $\frac{1}{2}, \frac{1}{2}, -\frac{2}{3}$   $\boxed{2}$

Q4

$$(d) x^2 + px + q = 0 \quad (\text{Roots } \alpha, \beta) \quad \text{--- (1)}$$

$$x^2 + p'x + q = 0 \quad (\text{Roots } k\alpha, \gamma \text{ say}) \quad \text{--- (2)}$$

$$(i) \text{ From (1) } \alpha\beta = q$$

$$\text{From (2) } k\alpha\gamma = q$$

$$\therefore \alpha\beta = k\alpha\gamma$$

$$\therefore \gamma = \frac{\beta}{k}$$

$$(ii) \text{ For (1) } \alpha + \beta = -p$$

$$\text{For (2) } k\alpha + \frac{\beta}{k} = -p'$$

$$\text{Substitute } \beta = -p - \alpha$$

$$\therefore k\alpha - \frac{p + \alpha}{k} = -p'$$

$$k^2\alpha - (p + \alpha) = -kp'$$

$$k^2\alpha - \alpha = p - kp'$$

$$\alpha(k^2 - 1) = p - kp'$$

$$\alpha = \frac{p - kp'}{k^2 - 1}$$

$$\text{Now } \beta = -p - \alpha$$

$$= -p - \left( \frac{p - kp'}{k^2 - 1} \right)$$

$$= \frac{-p(k^2 - 1) - (p - kp')}{k^2 - 1}$$

$$\therefore \beta = \frac{kp' - k^2p}{k^2 - 1}$$

$$(iii) \text{ Now } \alpha\beta = q$$

$$\therefore \frac{p - kp'}{k^2 - 1} \cdot \frac{k(p' - kp)}{k^2 - 1} = q$$

$$\therefore (p - kp')k(p' - kp) = (k^2 - 1)^2 q$$

$$\therefore k(kp - p')(kp' - p) = (k^2 - 1)^2 q$$

QED [1]

(iv) Expanding both sides and collecting terms on the left gives the quartic eq'n in  $k$ :

$$qk^4 - pp'k^3 + \dots = 0$$

$\therefore$  By sum of roots

$$\text{Sum of all } k's = \frac{pp'}{q}$$

QED

Q5(a) (i)

x x x x x x

Arrangements of girls =  $6!$

Choose 3 from 7 locations for boys =  ${}^7P_3$

$$\therefore \text{No of arrangements} = 6! \cdot {}^7P_3$$

$$= 151200$$

2

(ii)

(\*)

x  
x x  
x x

Arrangements of girls =  $5!$

Choose 3 from 6 locations for boys =  ${}^6P_3$

$$\therefore \text{No of arrangements} = 5! \cdot {}^6P_3$$

$$= 14400$$

1

(b) (i)  ${}^nC_2 = \frac{n(n-1)}{2}$

(ii)  $P(\text{odd, odd}) = \frac{\cancel{n+1}}{n} \cdot \frac{\frac{n-1}{2}}{n-1}$

$$= \frac{n+1}{2n} \cdot \frac{1}{2}$$

$$= \frac{n+1}{4n}$$

$$\frac{\frac{n}{2}-1}{2n-2}$$

$$= \frac{n-2}{4n-4}$$

2

(iii)  $P(\text{odd, odd}) = \frac{1}{2} \cdot \frac{\frac{n-2}{2}}{n-1}$

$$= \frac{1}{4} \cdot \frac{n-2}{n-1}$$

$$= \frac{n-2}{4n-4}$$

$$\frac{{}^mC_2}{{}^{2m}C_2}$$

$$= \frac{\frac{m(m-1)}{2}}{\frac{2m(2m-1)}{2}}$$

$$= \frac{m-1}{2m}$$

2

$$(c) \quad I_n = \int \sec^n x \, dx$$

$$= \int \sec^{n-2} x \cdot \sec^2 x \, dx$$

$$= \sec^{n-2} x \cdot \tan x$$

$$u = \sec^{n-2} x \quad v = \tan x$$

$$- \int (n-2) \sec^{n-3} x \cdot \sec x \tan x \cdot \tan x \, dx \quad u' = \sec^{n-3} x$$

$$v' = \sec^2 x$$

$$(n-2) \sec^{n-3} x \cdot \sec x \tan x$$

$$= \sec^{n-2} x \tan x$$

$$- (n-2) \int \sec^{n-2} x \cdot \tan^2 x \, dx$$

$$= \sec^{n-2} x \tan x$$

$$- (n-2) \int \sec^{n-2} x \cdot (\sec^2 x - 1) \, dx$$

$$= \sec^{n-2} x \tan x - (n-2) \left\{ \int \sec^n x \, dx - \int \sec^{n-2} x \, dx \right\}$$

$$\therefore I_n = \sec^{n-2} x \tan x - (n-2) \{ I_n - I_{n-2} \}$$

$$\therefore I_n + (n-2) I_n = \sec^{n-2} x \tan x + (n-2) I_{n-2}$$

$$\therefore I_n (n-1) = \sec^{n-2} x \tan x + (n-2) I_{n-2}$$

$$\therefore I_n = \frac{1}{n-1} \sec^{n-2} x \tan x + \frac{n-2}{n-1} I_{n-2}$$

2

(d) (i) If  $n = 4p$

$$(1+i^n)(1+i^{2n})$$

$$= (1+i^{4p})(1+i^{8p})$$

$$= 2 \cdot 2$$

$$= 4$$

If  $n = 4p+1$

$$(1+i^n)(1+i^{2n})$$

$$= (1+i^{4p+1})(1+i^{8p+2})$$

$$= (1+i)(1+i^2)$$

$$= (1+i)(1-i)$$

$$= 1$$

If  $n = 4p+2$

$$(1+i^n)(1+i^{2n})$$

$$= (1+i^{4p+2})(1+i^{8p+4})$$

$$= (1+i^2)(1+i)$$

$$= 0$$

4

If  $n = 4p+3$

$$(1+i^{4p+3})(1+i^{8p+6})$$

$$= (1-i)(1-i)$$

$$= 0$$

(ii) Function is  $f(n) = 365 + \frac{(1+i^n)(1+i^{2n})}{4}$

1

$$Q6 (a) \int_0^2 \sqrt{4+x^2} dx$$

$$\text{let } x = 2 \tan \theta$$

$$\therefore dx = 2 \sec^2 \theta d\theta$$

$$= \int_0^{\pi/4} \sqrt{4+4\tan^2 \theta} \cdot 2 \sec^2 \theta d\theta$$

$$\text{If } x=0, \theta=0$$

$$= \int_0^{\pi/4} 2 \sec \theta \cdot 2 \sec^2 \theta d\theta$$

$$\text{If } x=2, \theta = \frac{\pi}{4}$$

$$= 4 \int_0^{\pi/4} \sec^3 \theta d\theta$$

$$= 4 \left\{ \left[ \frac{1}{2} \sec \theta \tan \theta \right]_0^{\pi/4} + \frac{1}{2} \int_0^{\pi/4} \sec \theta d\theta \right\}$$

$$= 4 \left\{ \left[ \frac{1}{2} \sec \frac{\pi}{4} \cdot \tan \frac{\pi}{4} \right] - \left[ \frac{1}{2} \sec 0 \tan 0 \right] + \frac{1}{2} \left[ \ln (\sec \theta + \tan \theta) \right]_0^{\pi/4} \right\}$$

$$= 4 \left\{ \frac{1}{2} \cdot \sqrt{2} \cdot 1 - 0 + \frac{1}{2} \left[ \ln (\sec \frac{\pi}{4} + \tan \frac{\pi}{4}) \right] - \frac{1}{2} \ln (\sec 0 + \tan 0) \right\}$$

$$= 2\sqrt{2} + \frac{1}{2} \ln (\sqrt{2} + 1) - \frac{1}{2} \ln 1$$

$$= 2\sqrt{2} + \frac{1}{2} \ln (\sqrt{2} + 1)$$

2

$$(b) (i) \quad y = e^{-x^2} (x^2 + c)$$

$$\begin{aligned} y' &= (x^2 + c) \cdot e^{-x^2} \cdot -2x + e^{-x^2} \cdot 2x \\ &= 2x e^{-x^2} (-x^2 + c + 1) \end{aligned}$$

$$y' = 0 \quad \text{if } x = 0 \quad \text{or } x^2 + c + 1 = 0.$$

$$\therefore x^2 = \cancel{1-c} \quad 1-c$$

$$\therefore \cancel{1-c} \geq 0$$

$$\therefore c \leq 1, \text{ for zeros.}$$

$$\text{If } c = 1 \quad x = 0$$

If  $c < 1$  the curve has 3 stationary points.

otherwise the curve has 1 stationary point. 2

$$(ii) \text{ If } c = 2$$

$$\begin{aligned} y' &= -2x e^{-x^2} (x^2 + 1) \\ &= e^{-x^2} (-2x^3 - 2x) \end{aligned}$$

$$\begin{aligned} y'' &= e^{-x^2} \cdot (-6x^2 - 2) + \frac{d}{dx} (-2x) (x^2 + 1) \cdot e^{-x^2} - 2x \\ &= e^{-x^2} (-6x^2 - 2 + 4x^2(x^2 + 1)) \end{aligned}$$

$$= e^{-x^2} (-6x^2 - 2 + 4x^4 + 4x^2)$$

$$= e^{-x^2} (4x^4 - 2x^2 - 2)$$

$$= 2e^{-x^2} (2x^4 - x^2 - 1)$$

$$= 2e^{-x^2} (2x^2 + 1)(x^2 - 1)$$

$$\therefore x^2 = -\frac{1}{2} \quad \text{or } x^2 = 1$$

$$\therefore x = \pm \frac{1}{2} \quad \text{or } \pm 1$$

∴ Potential points of inflection are

$$(1, \frac{3}{2}) \text{ or } (-1, \frac{3}{2})$$

$$\text{If } x = 0.5 \quad y'' = +ve \times +ve \times -ve < 0$$

$$\text{If } x > 1.5 \quad y'' = +ve \times +ve \times +ve \geq 0$$

∴ Chge of concavity.

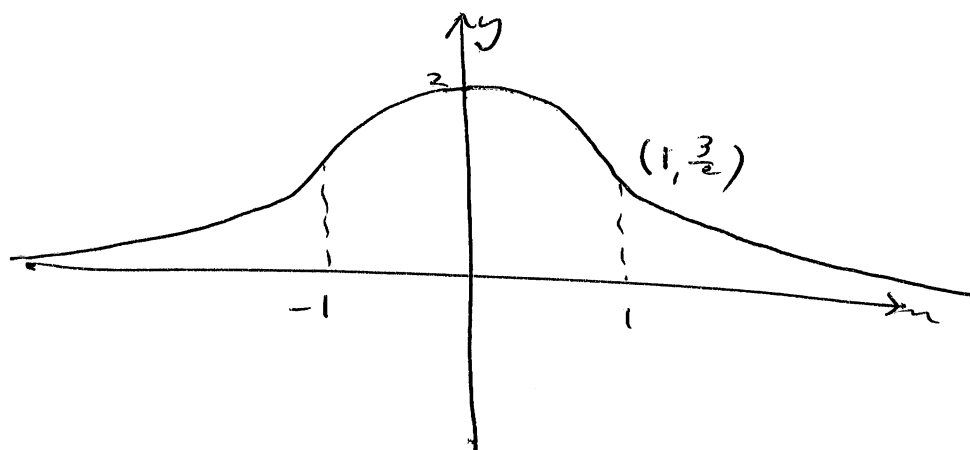
$(1, \frac{3}{2})$  is a point of inflection.

The function is an even function

∴  $(-1, \frac{3}{2})$  is also a point of inflection

(iii) When  $x=0$   $y=2$   ~~$y=2$~~

$$\text{As } x \rightarrow \infty \quad y \rightarrow 0$$



(v) (i)

$$\ddot{x} = -\left(10 + \frac{v^2}{10}\right)$$

↓  
so ↓ ↓  $\frac{v^2}{2}$

$$v \frac{dv}{dx} = -\frac{100 + v^2}{10}$$

~~$$\int \frac{v dv}{100 + v^2} = -\int \frac{dx}{10}$$~~

$$\int \frac{v dv}{100 + v^2} = -\int \frac{dx}{10}$$

$$\frac{1}{2} \ln(100 + v^2) = -\frac{1}{10} x + c$$

when  $x=0$ ,  $v=12$

$$\therefore \frac{1}{2} \ln 244 = c$$

$$\therefore -\frac{1}{10} x = \frac{1}{2} \ln(100 + v^2) - \frac{1}{2} \ln 244$$

$$\therefore \frac{1}{10} x = \frac{1}{2} \left\{ \ln(244) - \ln(100 + v^2) \right\}$$

$$\therefore \frac{1}{5} x = \ln \frac{244}{100 + v^2}$$

Max ht reached when  $v=0$

$$\therefore \frac{1}{5} x = \ln \frac{244}{100}$$

$$\therefore x = 5 \ln \frac{244}{100}$$

$$= 4.4899 \dots \text{ m}$$

2

$$(ii) \quad \frac{dr}{dt} = -\left(10 + \frac{r^2}{10}\right)$$

$$\therefore \frac{dr}{dt} = -\frac{100 + r^2}{10}$$

$$\therefore \int \frac{dr}{100 + r^2} = \int -\frac{1}{10} dt.$$

$$\therefore \frac{1}{10} \tan^{-1} \frac{r}{10} = -\frac{1}{10} t + c.$$

When  $t=0$ ,  $r=12$

$$\frac{1}{10} \tan^{-1} \frac{12}{10} = c.$$

$$\therefore \frac{1}{10} \tan^{-1} \frac{r}{10} = -\frac{1}{10} t + \frac{1}{10} \tan^{-1} \frac{12}{10}.$$

$$\therefore \tan^{-1} \frac{r}{10} = -t + \tan^{-1} \frac{12}{10}.$$

$$\therefore t = \tan^{-1} \frac{12}{10} - \tan^{-1} \frac{r}{10}.$$

If  $r=0$ ,  $t = \tan^{-1} \frac{12}{10}$   
 $= 0.8760 \dots s$

2

~~scribble~~

(iii)

$$\uparrow \frac{v^2}{2}$$

$$\ddot{x} = 10 - \frac{v^2}{10}$$

$$\downarrow 10$$

$$\frac{v dv}{dx} = \frac{100 - v^2}{10}$$

$$\therefore \int \frac{v dv}{100 - v^2} = \int \frac{1}{10} dx$$

$$-\frac{1}{2} \ln(100 - v^2) = \frac{1}{10} x + c$$

$$\text{When } x=0, v=0: -\frac{1}{2} \ln 100 = c$$

$$\therefore -\frac{1}{2} \ln(100 - v^2) = \frac{1}{10} x - \frac{1}{2} \ln 100$$

$$\frac{1}{10} x = \frac{1}{2} (\ln 100 - \ln(100 - v^2))$$

$$x = 5 \left( \ln \frac{100}{100 - v^2} \right)$$

$$\text{If } x = 5 \ln \frac{244}{100} :$$

$$5 \ln \frac{244}{100} = 5 \ln \frac{100}{100 - v^2}$$

$$\therefore \frac{244}{100} = \frac{100}{100 - v^2}$$

$$\therefore 100 - v^2 = \frac{10000}{244}$$

$$v^2 = 100 - \frac{10000}{244}$$

$$v^2 = \frac{24400 - 10000}{244}$$

$$= \frac{14400}{244}$$

$$\therefore v = \frac{120}{\sqrt{244}}$$

$$= 27.6222 \text{ m/s}$$

2